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- **Problem Statement ID – SIH25183**
- **Problem Statement Title- AI enabled Cyber incident & Safety web portal for Defence.**
- **Theme- Blockchain & Cybersecurity**
- **PS Category- Software**
- **Team ID- 73338**
- **Team Name : Six-Strangers**

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AI Powered Defence Cyber Safety Portal

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❖ Proposed Solution (Describe your Idea/Solution/Prototype)

Objective: To develop an **AI-driven Defence Cyber Portal** that empowers serving personnel, veterans, and families to **detect, analyze, and report cyber threats**—ensuring rapid triage and **priority escalation to CERT-Army**.

Dataset Source: **Curated datasets** comprising **phishing URLs, forged video samples, morphed image datasets, and APK malware binaries**, aggregated from Kaggle, PhishTank, VirusTotal, and CICMalDroid repositories.

Data Segmentation: Divides collected threat **datasets into 19 segments**, each trained with **AI/ML models** specialized for **NLP, CNN, and Random Forest**, ensuring balanced and efficient threat detection.

Threat Extractor: **Extracts and normalizes** multi-format defence threat features, **including APK indicators, domain names, byte patterns, entropy, structural signatures**, creating a standardized feature vector for AI/ML-based threat detection.

Complaint Handling: Securely ingests **multi-format complaints** and **classifies them via AI/ML** and storing all incidents with **encryption** and **role-based access control.**

Uniqueness:

- **Dynammic model:** AI models continuously update from **emerging and zero-day threats**, adapting to **new attack patterns.**
- **Generative AI Guidance:** uses Gen AI to provide automated **mitigation suggestions** and **step by step guidance**

List of Algorithm used:

BERT LSTM CNN
Random Forest
XGBoostLight GBM RNN
GRU ResNet
EfficientNet YOLO
SVM K-Means
SMTP RSA-256 RBAC

Product status

Prototype: completion status- ***85% completed.***

Demo Link

Prototype Demo & Explanation: ***VIDEO LINK CLICK HERE***

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TECHNICAL APPROACH

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DATA PREPARATION : 1

Dataset: **Debian Malware Dataset** for APKs,**PhishTank Datasets** for Phishing URLs, **CASIA v2** for Image Tampering, **Deepfake Detection Challenge Dataset** for video Analysis.

Preprocessing: Cleans and standardizes inputs by **removing noise, decoding URLs, resizing media**, and converting data into model-readable numerical vectors

FEATURE EXTRACTION: 2

Byte Pattern Analysis: Counts and **normalizes byte frequencies** in APKs.

URL Pattern Analysis: Extracts **and encodes domain structure**

Image Pattern Analysis: Extracts **texture, color, and edge inconsistencies**

Video Frame Analysis: Analyzes **frame-by-frame temporal patterns and visual artifacts.**

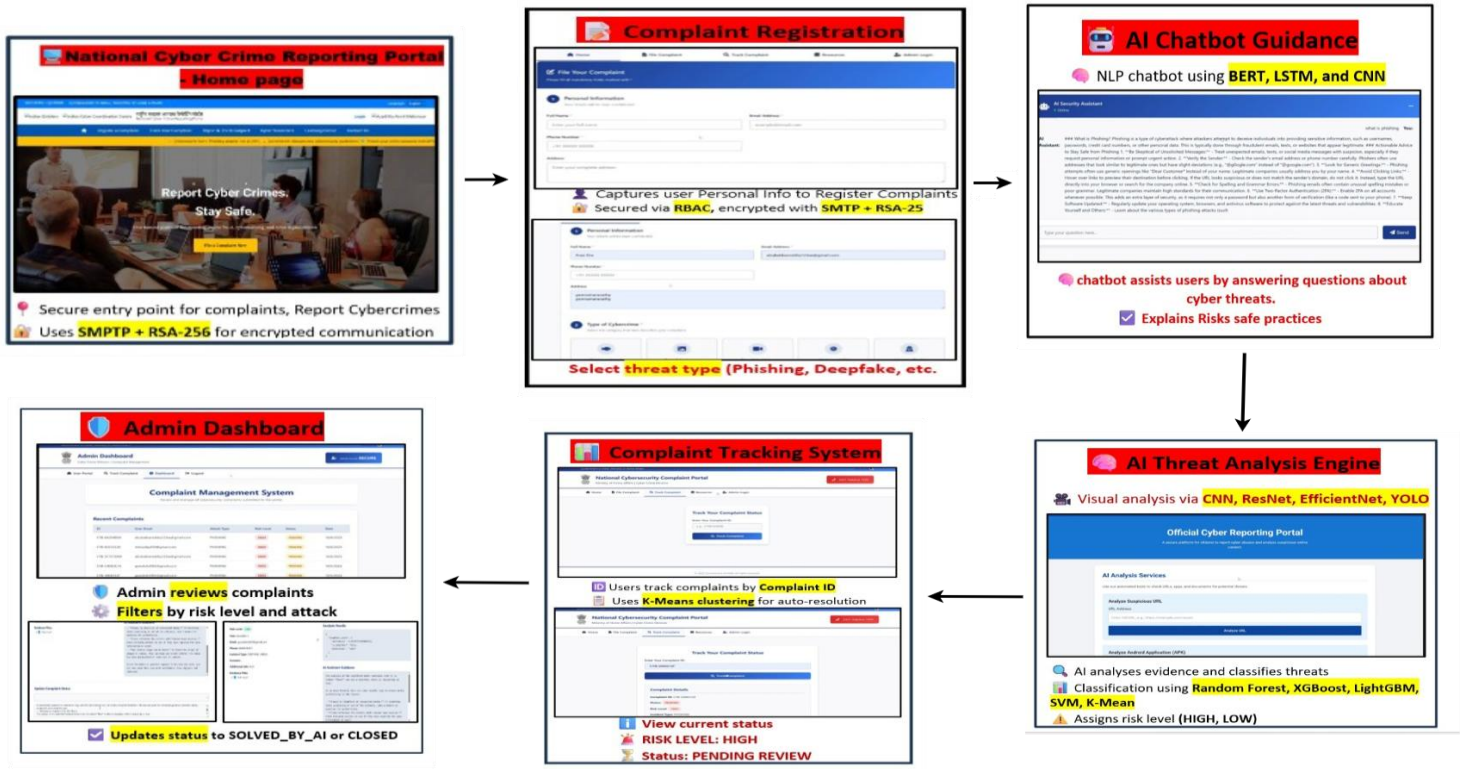
Threat Detector: Combines **byte, domain, and visual features into a unified fixed-length vector**, creating a multi-dimensional threat signature for AI-based classification

MODEL TRAINING:- 3

BERT & LSTM Model: Learns **contextual patterns from phishing text and URLs** and analyzes sequential behavior.

CNN Model: **Processes image and video data** to identify morphing, tampering, and deepfake threats.

Random Forest: Learns **malware patterns from APK features**



CERT Portal and defence suggestion : 4

- **Gen AI Guidance:** Provides **real-time threat mitigation suggestions using LLMs and RAG**.
- **Evidence Handling:** **Data Chain of Custody** with **secure database logging**
- **Real-Time Notification:** Continuous, **multi-channel alerting** triggered by incident severity and delivered via **email and secure webhooks**.

WEB INTERFACE: 5

Provides **a web-based interface** for defence members, veterans, and families to submit cyber complaints.

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FEASIBILITY AND VIABILITY

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Model Efficiency: Uses **Random Forest** for text/URLs and **CNN** for images/videos to **handle diverse cyber threats** effectively.

Dynamic Loading: Handle Multi-format complaints (text, URLs, files, images, videos) sequentially to effectively handle resources.

Efficient Resource Utilization: Minimizes **unnecessary computation** and ensures optimal use of resources for faster threat detection.

Cross-Platform Compatibility: Developed as a **secure web portal**, which can be deployed on **defence intranets** as well as can be accessed on normal browsers.

Scalable Solution: Easily adaptable to **growing datasets** and evolving cyber threats targeting defence personnel.

Resource Optimization: Processes complaints with **limited computational overhead**, reducing operational costs..

Modular Approach: Modular design enables **simple updates and AI model upgrades**, which enable effortless defence integration.

Versatility: It can detect **phishing, malware, fraud, espionage, and OPSEC** threats and can be used in a variety of defence cyber cases.

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IMPACT AND BENEFITS

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IMPACTS

Enhanced Cyber Safety

Increased Cyber Safety Defence personnel, and families that will be **safeguarded** against **phishing, malware, fraud, espionage.**

Faster Threat Detection

AI ensures **real-time identification** of suspicious emails, files, and links, **reducing risk of late response.**

Timely Response by CERT-Army

Provides **risk-ranked dashboards** that help CERT-Army act faster and more effectively.

Security Protection

Stops the leakage of **sensitive defence information** and the threat of espionage.

BENEFITS

User Awareness & Guidance

Delivers instant alerts and **step-by-step mitigation instructions** to users for safe practices.

Field Accessibility

Web based portal ensures **easy complaint submission** even in remote or field conditions.

User-Friendly Interface

Simple and **intuitive** design with clear navigation, making it easy for defence personnel and families to **submit complaints** and follow mitigation steps without **technical expertise**.

Resource Efficiency:

Reduces incident handling time, **prevents major cyber damages.**

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RESEARCH AND REFERENCES

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